Mansoura University
Faculty of Computers and Information
Department of Computer Science

PROJECT PROPOSAL

Ninja

Submitted by:

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PROJECT ABSTRACT:

What is competitive programming?

-Competitive programming involves programmers competing against each other to solve programming questions in a limited amount of time. These events are usually held over the Internet or a local network.

-Contestants solve logical or mathematical programming questions in the coding language of their choice. The number of questions can vary depending on the contest. The results are judged on the number of questions solved, time spent solving the problems, and other factors, like execution time, program size, etc.

- ~~There are many places you can compete. Some of these places will also allow you to~~ sign up and take practice challenges like Codeforces, LeetCode, HackerRank, etc.

What is Ninja?

- Ninja is an android app focuses mainly on helping competitive programmers to improve their performance by visualizing some statistics about the user, view information about upcoming contests and compare his performance with other contestants.

PROJECT OBJECTIVES:

App features: -

-Single User Analytics

- -Verdict's chart.
- -Language's chart.
- -Tags chart.
- -Levels chart.
- -Total tried problems count.
- -Total solved problems count.
- -Average and max attempts.
- -Count of problems solved with one submission.
- -Max AC for a single problem.
- -List of unsolved problems.

-Comparison between two users

- -Current, max and min rating.
- -Number of contests.
- -Best and worst position in contest.
- -Max positive and negative rating change.
- -Compared rating timeline.
- -Total tried problem count compared.
- -Total solved problem count compared.
- -Average and max attempts compared.
- -Count of problems solved with one submission compare.
- -Max AC for a single problem compared.
- -Tags compared.
- -Levels compared.

WHO IS THE PROJECT **COMPETITIVE**? AND HOW WILL YOUR PROJECT BE **DIFFERENT**?

- PROJECT COMPETITIVE: - **Codeforces Visualizer (website)**
- My project includes the same functionality but as a mobile app as it will be easier to use for most users and it supports other platforms not only codeforces.

TOOLS, HARDWARE, AND SOFTWARE RESOURCES:

Tools: -

- Java.
- Kotlin.
- Codeforces API.
- Kontests API.

Software: -

- Android studio.
- Photoshop

SCHEDULING PHASES:

From	To	Activity
23/4	24/4	Requirements analysis
25/4	31/4	Prototyping and Design
1/5	14/5	Development
15/5	21/5	Testing
22/5	28/5	Deployment and maintenance

REFERENCES:

- **Kontests API**
- **Codeforces API**